

ComicCraft – Project Documentation

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Team Size: 5

Project Name: ComicCraft – AI Comic Story Creator Using Gemini Models
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Abstract

ComicCraft is an AI-assisted comic-story creator that transforms a short story idea into a structured five-panel narrative. The documented solution uses a lightweight Python/Streamlit interface and Google's Gemini model to generate scene descriptions, character actions, dialogue/captions and image prompts. Character consistency is encouraged through prompt design. The current documented scope is text generation rather than automatic comic artwork.

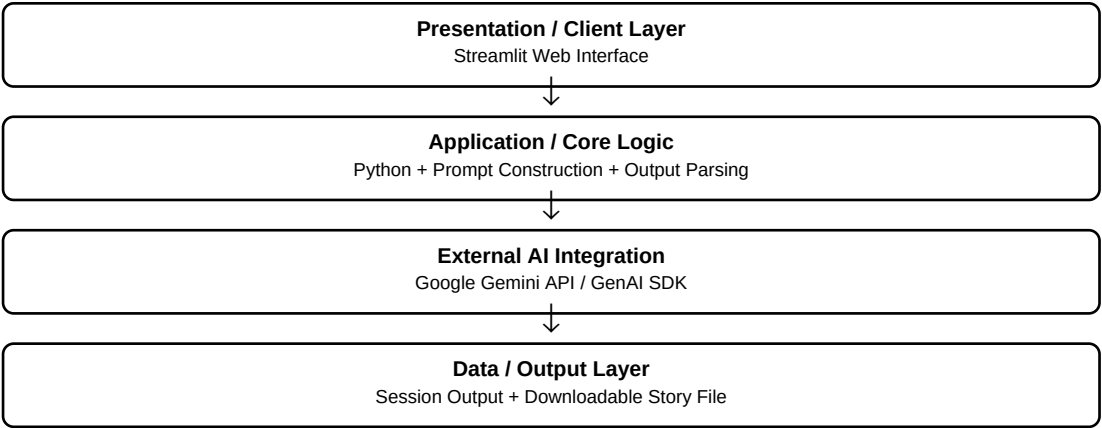
Problem Statement

Many aspiring storytellers have ideas but lack the time or skills to structure those ideas into sequential comic panels. Traditional comic planning requires narrative writing, sequencing, dialogue and visual planning. General-purpose AI writing can also return prose that requires manual conversion into panel structure.

Objectives

- Generate a complete five-panel comic story from a short idea.
- Use Gemini for generative narrative creation.
- Separate scene, action, dialogue and image-prompt content.
- Maintain character consistency through prompt engineering.
- Provide downloadable output.
- Keep API credentials outside source code.

System Architecture



Technology Stack

Layer	Technology
Interface	Streamlit
Language	Python

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Generative AI	Google Gemini
SDK	Google GenAI SDK
Configuration	python-dotenv
Version Control	Git / GitHub

Core Features

- Five-panel story generation
- Scene generation
- Character actions
- Dialogue/captions
- Image prompts
- Character consistency
- Downloadable output
- Simple user interface

Testing

The supplied report documents functional, output-structure, character-consistency and error-handling tests. Eight test cases are recorded as Pass, including valid/empty input, missing API key, download, repeated stories, consistency and long input handling.

Limitations and Future Enhancements

- No automatic comic artwork in the documented current scope.
- LLM output can vary in quality and consistency.
- No persistent story history or user accounts.
- Fixed five-panel format.
- Future work includes image generation, variable panel counts, accounts/history, richer export formats, style/genre controls, collaboration and hosted deployment.

Submission consistency note

The supplied source ZIP contains a FastAPI/Diffusers-oriented prototype while the supplied report documents a Streamlit application. The two should be reconciled before final execution or evaluation.